Homework 3 Report

After reading the assignment and downloading the datasets provided by the professor, I started by working on the dimensional model required for the data warehouse.

Part 1 – Data Warehouse Modeling

In this phase, I analyzed the available datasets and identified the business process as the analysis of financial transactions performed during 2024.

I defined the grain of the fact table as one transaction from the account statement file. Based on this, I designed a star schema composed of one fact table (Fact_Transactions) and four dimension tables: Dim_Time, Dim_Symbol, Dim_Geography, and Dim_TransactionType.

After identifying the dimensions, I defined the hierarchies for Time, Geography, and Symbol and then created the final star schema using dbdiagram.io. I also defined the surrogate keys and the relationships between the fact table and the dimensions.

Part 2 – ETL and Analytical Queries

For this phase, I loaded the three CSV files into Jupyter Notebook using pandas.

Before creating the dimensional model, I performed some basic data quality checks. I checked for missing values, removed unnecessary columns, verified that all transaction symbols existed in the symbols dataset, and verified that the countries could be matched correctly with the country dataset.

After cleaning the data, I converted the Date column into datetime format and created the dimension tables required by the star schema. I then generated the surrogate keys and built the Fact_Transactions table by linking the dimensions through foreign keys.

Once the ETL process was completed, I answered the analytical questions required by the assignment. For each query, I produced the requested results and added a short interpretation of the output.

Part 3 – Streamlit Dashboard

In the final phase, I created the Streamlit dashboard required by the assignment.

The dashboard focuses on Time Analysis and includes a date range filter that allows users to select a period between January 1st and December 31st, 2024. All charts automatically update according to the selected dates.

The dashboard contains a line chart showing the total number of transactions over time, a bar chart with the top three traded symbols, a bar chart with the top five sectors by transaction count, and a bar chart with the top five industries by transaction count.

During this phase, I also organized the project files and connected the dashboard to the CSV files generated during the ETL process.

Final Considerations

This homework allowed me to apply the concepts studied during the course, including dimensional modeling, ETL processes, analytical queries, and dashboard development. The final result is a complete workflow that starts from raw data and ends with an interactive dashboard for data analysis.